

Problem 1. Fibonacci Divisibility

(a)

Proof by strong induction. Let $P(n) := F_{n+a} \equiv F_n \cdot F_{a+1} \pmod{F_a}$, now to prove $p(n)$ for $n \geq 0$ by strong induction.

Base Case:

$$P(0) = F_a \equiv F_0 \cdot F_{a+1} \equiv 0 \pmod{F_a}.$$

$$P(1) = F_{a+1} \equiv F_1 \cdot F_{a+1} \equiv 1 \cdot F_{a+1} \equiv F_{a+1} \pmod{F_a}$$

Therefore, both cases hold.

Inductive step: For strong induction on n assume $P(n-1)$ and $P(n-2)$ holds for $n \geq 2$. So,

$$F_{(n-1)+a} \equiv F_{n-1} \cdot F_{a+1} \pmod{F_a} \quad (1)$$

$$F_{(n-2)+a} \equiv F_{n-2} \cdot F_{a+1} \pmod{F_a} \quad (2)$$

Now by property of the congruence on equation (1) and (2),

$$F_{(n-1)+a} + F_{(n-2)+a} \equiv F_{n-1} \cdot F_{a+1} + F_{n-2} \cdot F_{a+1} \pmod{F_a}$$

$$F_{n+a} \equiv (F_{n-1} + F_{n-2}) \cdot F_{a+1} \pmod{F_a}$$

$$F_{n+a} \equiv F_n \cdot F_{a+1} \pmod{F_a}$$

Which implies $P(n)$ is true. Therefore by strong induction hypothesis $P(n)$ is true for all $n \geq 0$. □

(b)

Proof by induction. Induction hypothesis Assume $P(n)$ is true, i.e.,

$$F_{na} \equiv 0 \pmod{F_a}.$$

Base Case: $P(0) = F_0 \equiv 0 \pmod{F_a}$

Base case holds. We now prove $P(n+1)$, namely,

$$F_{(n+1)a} \equiv 0 \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \pmod{F_a}.$$

Using part (a) with n replaced by na ,

$$F_{na+a} \equiv F_{na} F_{a+1} \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \cdot F_{a+1} \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \pmod{F_a}.$$

Which proves $P(n+1)$ holds and by induction hypothesis $P(n)$ is true for $n \geq 0$. □

Now if $a|b$ then $\exists k \in \mathbb{N}$ such that $a * k = b$. Now by $P(k)$, $F_{ak} = F_b \equiv 0 \pmod{F_a}$ which implies $F_a|F_b$.

Problem 2. Computing Modular Inverses

(a) First using Euclid's algorithm:

$$\begin{aligned} \gcd(43, 15) &= \gcd(15, 13) & 13 &= 43 - 2 \cdot 15 \\ &= \gcd(13, 2) & 2 &= 15 - 1 \cdot 13 \\ &= \gcd(2, 1) & 1 &= 13 - 6 \cdot 2 \end{aligned}$$

now doing backwards substitution (Pulverizer):

Substitute 2 (from $2 = 15 - 1 \cdot 13$)

$$\begin{aligned} 1 &= 13 - 6 \cdot (15 - 1 \cdot 13) \\ 1 &= 7 \cdot 13 - 6 \cdot 15 \end{aligned}$$

Substitute 13 (from $13 = 43 - 2 \cdot 15$)

$$\begin{aligned} 1 &= 7 \cdot (43 - 2 \cdot 15) - 6 \cdot 15 \\ 1 &= 7 \cdot 43 - 14 \cdot 15 - 6 \cdot 15 \\ 1 &= 7 \cdot 43 - 20 \cdot 15 \end{aligned}$$

The coefficient of 15 is -20 Therefore, the Inverses would be $43 - 20 = 23$. So $23 \cdot 15 \equiv 1 \pmod{43}$.

(b) From Fermat's Little theorem we know that $15^{42} \equiv 1 \pmod{43}$ and now multiplying both side with 15^{-1} we get

$$15^{41} \equiv 15^{-1} \pmod{43}$$

Now to simplify 15^{41} we calculate the smaller powers of 15:

$$\begin{aligned} 15^1 &\equiv 15 \pmod{43} \\ 15^2 &= 225 \equiv 10 \pmod{43} \\ 15^4 &\equiv 100 \equiv 14 \pmod{43} \\ 15^8 &\equiv 196 \equiv 24 \pmod{43} \\ 15^{16} &\equiv 576 \equiv 17 \pmod{43} \\ 15^{32} &\equiv 289 \equiv 31 \pmod{43} \end{aligned}$$

Now $15^{41} = 15^{32} \cdot 15^8 \cdot 15$ Therefore, $15^{-1} \equiv 31 \cdot 24 \cdot 15 \equiv 23 \pmod{43}$ Meaning Inverses of 15 mod 43 is 23.

Problem 3. Primality Testing

(a)

Proof by Contradiction. Assume an integer $1 \leq a \leq n-1$ is fermat's witness of some prime number n . since a is fermat's witness $a^{n-1} \not\equiv 1 \pmod{n}$. Also since n is prime and a is in range $[1, n-1]$, Therefore a is not multiple of n so by fermat's little theorem $a^{n-1} \equiv 1 \pmod{n}$ which is a contradiction. $\square$

(b)

Proof. Assume p is a prime and $x^2 \equiv_p 1$ by the definition of congruence $p \mid x^2 - 1$

$$p \mid (x-1)(x+1)$$

Since p is prime (by Lemma 9.4.2) $p \mid (x-1)$ and $p \mid (x+1)$ so by definition of congruence $x \equiv_n -1$ and $x \equiv_n 1$. $\square$

(c)

Proof by contradiction. Assume, for the sake of contradiction, that the algorithm returns 'composite' for some prime number n . So at least one of the condition from the algorithm must hold.

- $x_e \not\equiv_n 1$
- Consecutive terms (x_i, x_{i+1}) where $x_{i+1} \equiv_n 1$ but $x_i \not\equiv_n \pm 1$.

Suppose the first condition holds then $x_e = a^{n-1} \not\equiv_n 1$. Now Since n is prime and $1 \leq a \leq n-1$ and Fermat's Little Theorem implies that $a^{n-1} \equiv_n 1$, which is a contradiction.

First condition led to contradiction, So second condition must be true. So $x_{i+1} = x_i^2 \equiv_n 1$ but $x_i \not\equiv_n \pm 1$. Since n is prime, by lemma from part (b), given condition yields contradiction.

Both of the condition lead to contradiction meaning our assumption must be wrong so n must indeed be composite. $\square$

Problem 4. Pulverizer State Machine

(a) For some arbitrary state (x, y, s, t, u, v) say Pres1, Pres2, Pres3 are true and we need to prove that Pres1, Pres2, Pres3 are true for the next state (x', y', s', t', u', v') .

By the definition of transition in given state machine we have $x' = y$, $y' = (x \text{ rem } y)$, $s' = u$, $t' = v$, $u' = s - qu$ and $v' = t - qv$ where $x = qy + (x \text{ rem } y)$.

- **Pres1:** $\gcd(x', y') = \gcd(a, b)$ we know that $\gcd(x', y') = \gcd(y, x \text{ rem } y) = \gcd(x, y) = \gcd(a, b)$ which proves Pres1 is preserved.
- **Pres2:** $s'a + t'b = x'$, using the definition of s', t', x' we have $s'a + t'b = ua + vb = y = x'$ which proves Pres2 is preserved.
- **Pres3:** $u'a + v'b = y'$, using the definition of u', v', y' we have

$$\begin{aligned}
 u'a + v'b &= (s - qu)a + (t - qv)b \\
 &= (sa + tb) - q(ua + vb) \\
 &= x - q(y) \\
 &= (x \text{ rem } y) \\
 &= y'
 \end{aligned}$$

which proves Pres3 is preserved.